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Research article

A hybrid ultrasonic strategy for super duplex stainless steel welding: Coupling arc induced acoustics and mechanical vibration to refine grains and balance phases

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ABSTRACT

Non-equilibrium solidification in arc-based welding and melting-based additive manufacturing often leads to coarse microstructures and segregation under steep thermal gradients. Although ultrasonic assistance has been explored as an additional process control route, its effectiveness in high-density metallic melts is frequently limited by strong acoustic attenuation, elevated cavitation thresholds, and stage-dependent action. This work proposes a hybrid ultrasonic strategy that couples ultrasonic frequency pulsed arc with external mechanical vibration to overcome these limitations. The fundamental principle lies in the phased co-regulation of molten pool hydrodynamics and solidification behavior via the integration of pulsed arc and mechanical ultrasonic vibration. Specifically, the pulsed arc can tailor the molten pool thermal state and thereby enhance the effectiveness of externally applied mechanical ultrasonic, enabling stronger solidification-stage regulation in dense metallic melts. This strategy is demonstrated in autogenous tungsten inert gas welding of 2507 super duplex stainless steel. Compared with the conventional method, the hybrid strategy achieved significant microstructural refinement: the average ferrite grain size was reduced from 143.4 μm to 70.2 μm, while the austenite area fraction increased from 6.9% to a more balanced 20.0%. These microstructural evolutions led to a simultaneous 16% increase in tensile strength and a remarkable 97.4% enhancement in ductility. These findings provide a physically grounded, transferable framework to mitigate coarse microstructure and phase imbalances in phase-sensitive alloys. This strategy can inspire ultrasonic-assisted processing for various heat sources and structural alloys in welding and additive manufacturing, addressing challenges including microstructural tailoring and precision phase regulation.

1. Introduction

Metal processing routes based on localized melting, such as fusion welding and melt-based additive manufacturing, face intrinsic challenges associated with non-equilibrium solidification. Under steep thermal gradients, the solidification front tends to develop coarse columnar dendrites and severe elemental segregation, which degrade the structural reliability and service performance of metallic components. This issue is particularly critical in high-performance alloys, whose properties are highly sensitive to the evolution of grain morphology and phase constitution during non-equilibrium solidification. Super duplex stainless steels (SDSS) are widely used in marine, energy, and chemical applications [1,2], often exhibit coarse grains and imbalanced ferrite (α)/austenite (γ) ratios after arc-based welding and deposition, which may deteriorate mechanical integrity and corrosion performance through microstructural heterogeneity [3,4]. Comparable solidification-induced coarsening has also been reported in titanium and nickel-based alloys [5,6]. More broadly, the instability of process–microstructure coupling remains a persistent bottleneck in fusion welding and melt-based additive manufacturing, limiting the attainable performance of structural alloys. Therefore, developing transferable strategies for regulating non-equilibrium solidification remains a central challenge in advanced metal processing.

Conventional efforts focus on thermal control and alloy design to mitigate non-equilibrium-solidification-induced microstructural heterogeneity in SDSS welding and additive manufacturing. Heat-input

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regulation strategies, such as pulsed arc [7], cold metal transfer (CMT) [8], and hot-wire techniques [9], can improve dimensional precision, suppress excessive ferrite, and enhance tensile strength [10,11]. However, their effectiveness is often strongly dependent on processing window, and insufficient heat input may increase susceptibility to porosity and lack of fusion [12,13]. Conversely, approaches that prolong the thermal cycle can promote austenite formation, but frequently at the cost of grain coarsening and toughness degradation [3,14]. Composition-based regulation, including Ni and N alloying [15,16] and tailored flux-cored wire [9], can rebalance the α/γ phase ratio, yet may introduce segregation, nitride precipitation [17,18], and increased process complexity. More fundamentally, these strategies mainly modify global process variables, which simultaneously influence molten pool flow, dendrite growth, solute redistribution, and phase transformation. Because these mechanisms are strongly coupled, improving one aspect often compromises another. Consequently, stage-specific regulation from solidification to post-solidification transformation remains difficult. Moreover, simultaneous optimization of grain morphology, phase balance, and interphase boundary characteristics is also challenging in SDSS and other alloys.

Ultrasonic assistance, as an additional process control, offers a promising physical-field approach to regulate molten pool dynamics and solidification through acoustic streaming and cavitation [19,20]. Two primary ultrasonic modes have been investigated in fusion welding and wire-based directed energy deposition: externally applied mechanical ultrasonic vibrations using solid transducers, and arc-induced acoustic excitation generated via high-frequency current modulation. In previous studies, the two ultrasonic modes have shown different roles in process regulation. Mechanical ultrasonic vibration has proven effective in refining the microstructure of light alloys such as aluminum alloys [21]. However, in high-density metallic melts its action is often spatially localized and stage-limited, because cavitation activation is more difficult and ultrasonic energy attenuates strongly, especially in mushy regions and alloys with narrow solidification ranges [22,23]. Arc-induced acoustic excitation is inherently coupled to the heat source and can enhance molten pool convection through electromagnetic-acoustic agitation and Marangoni-flow modification [24,25], but its acoustic intensity is typically limited and decays rapidly in dense melts, restricting its direct influence on solidification. Therefore, the key challenge is not simply whether ultrasonic assistance can be applied to SDSS, but how ultrasonic assistance should be configured to provide complementary control at different stages, from molten pool flow and solidification behavior to phase transformation. A schematic summary of the motivation and the proposed strategy is shown in Fig. 1.

Motivated by the stage-dependent limitations of single-mode ultrasonics in high-density metallic melts, this work proposes a hybrid ultrasonic strategy that couples ultrasonic-frequency pulsed arc with externally applied mechanical ultrasonic vibration. The pulsed arc can regulate heat source morphology and molten pool flow, whereas mechanical vibration is intended to strengthen acoustically induced effects during solidification. However, the extent of improvement in microstructure and mechanical properties enabled by the hybrid strategy, and whether the two acoustic fields can generate synergy beyond simple superposition, remain insufficiently evidenced and mechanistically explained.

To address these gaps, autogenous tungsten inert gas (TIG) welding of 2507 SDSS is selected as a stringent case study. The hybrid strategy is evaluated against conventional TIG welding and each single ultrasonic mode. The evaluation combines process behavior, microstructure evolution, and mechanical properties. The results show that the hybrid strategy can enhance weld penetration, refine the solidification microstructure, and promote a more favorable phase interface, leading to improved mechanical properties, and providing indications of synergistic interactions between the two acoustic fields. Specifically, pulsed arc can modify the molten pool state and enhance its responsiveness to mechanically induced acoustic effects, thereby enabling more effective solidification control.

Overall, these findings not only support the potential of hybrid ultrasonic strategy in SDSS welding but also provide transferable mechanistic rationale for this strategy. The hybrid ultrasonic strategy and the acoustic response control idea in this study can provide guidance and technical inspirations for welding and additive manufacturing processes targeting various heat sources and other structural alloys. This is conducive to addressing key challenges such as the precise design of microstructures and the refined control of phases in multiphase alloys.

2. Materials, process design and experimental methods

2.1. Process design and rationale of the hybrid ultrasonic strategy

The hybrid ultrasonic TIG strategy was proposed to regulate molten pool flow, solidification, and ferrite–austenite phase transformation during the TIG welding of 2507 SDSS. The strategy is not the simple addition of two ultrasonic sources, but the functional coupling between a heat-source-coupled pulsed arc and a contact-coupled mechanical vibration. Specifically, the ultrasonic-frequency pulsed arc was designed to modify the arc force and heat flux distribution, constrict the arc, and alter the thermal–flow state of the molten pool. The mechanically introduced ultrasonic vibration was intended to provide ultrasonic input through the welded plate and strengthen acoustic effects during solidification. Fig. 2a shows the main process regions affected by the two ultrasonic inputs. In addition, their coupled action is expected to increase the effective acoustic disturbance in the weld region, especially near the molten pool and solidification front.

Based on this strategy, the experimental system integrated an ultrasonic-frequency pulsed welding power source with a longitudinally vibrating ultrasonic roller, as shown in Fig. 2b. The ultrasonic-frequency pulsed current from the welding power source exhibited an approximately square-wave waveform. At the same time, the mechanically driven ultrasonic roller was placed in contact with the plate to introduce an additional high-frequency mechanical vibration. Thus, the two ultrasonic inputs were introduced through different physical routes: one through the arc and molten pool, and the other through the workpiece.

The mechanical ultrasonic vibration was introduced using a longitudinal ultrasonic roller assembly consisting of an ultrasonic generator (HC-CS-2000E, Jiazhen Ultrasonic Technology, Hangzhou, China), with a rated electrical power 2000 W, a piezoelectric transducer with a center frequency of 35 kHz, and a titanium alloy vibrating roller with outer diameter 96 mm, width 10 mm. The vibration was applied normal to the plate, as shown in Fig. 2b. The working frequency was set to 35 kHz with a frequency stability of approximately ± 0.5 kHz during welding. The vibration amplitude at the roller surface was 12 μm under no-load condition. The ultrasonic power reported in Table 1 refers to the electrical output power of the generator. The operating power was set to 800 W, which was selected from preliminary power-screening trials as a suitable condition that provided stable welding and an evident ultrasonic effect.

To ensure stable mechanical coupling, the roller was positioned slightly lower than the torch, by approximately 6 mm, with an arc length of 4 mm, resulting in controlled geometric interference between the roller and the workpiece surface. Under this configuration, the normal contact was mainly provided by the self-weight of the roller assembly (≈5.5 kg, corresponding to a static load of ≈54 N) and the high stiffness of the mounting fixture. The roller operated in a free-rolling mode driven by the travel motion, and its rotational speed was not actively controlled.

Given the thin plate geometry and the long wavelength at 35 kHz in stainless steel, the mechanical input is expected to manifest primarily as plate-scale structural vibration with guided-wave characteristics, similar to previous work [28,29]. Accordingly, the ultrasonic vibration herein refers to the contact-coupled high-frequency oscillatory input transmitted to the weld zone. The roller was positioned behind the

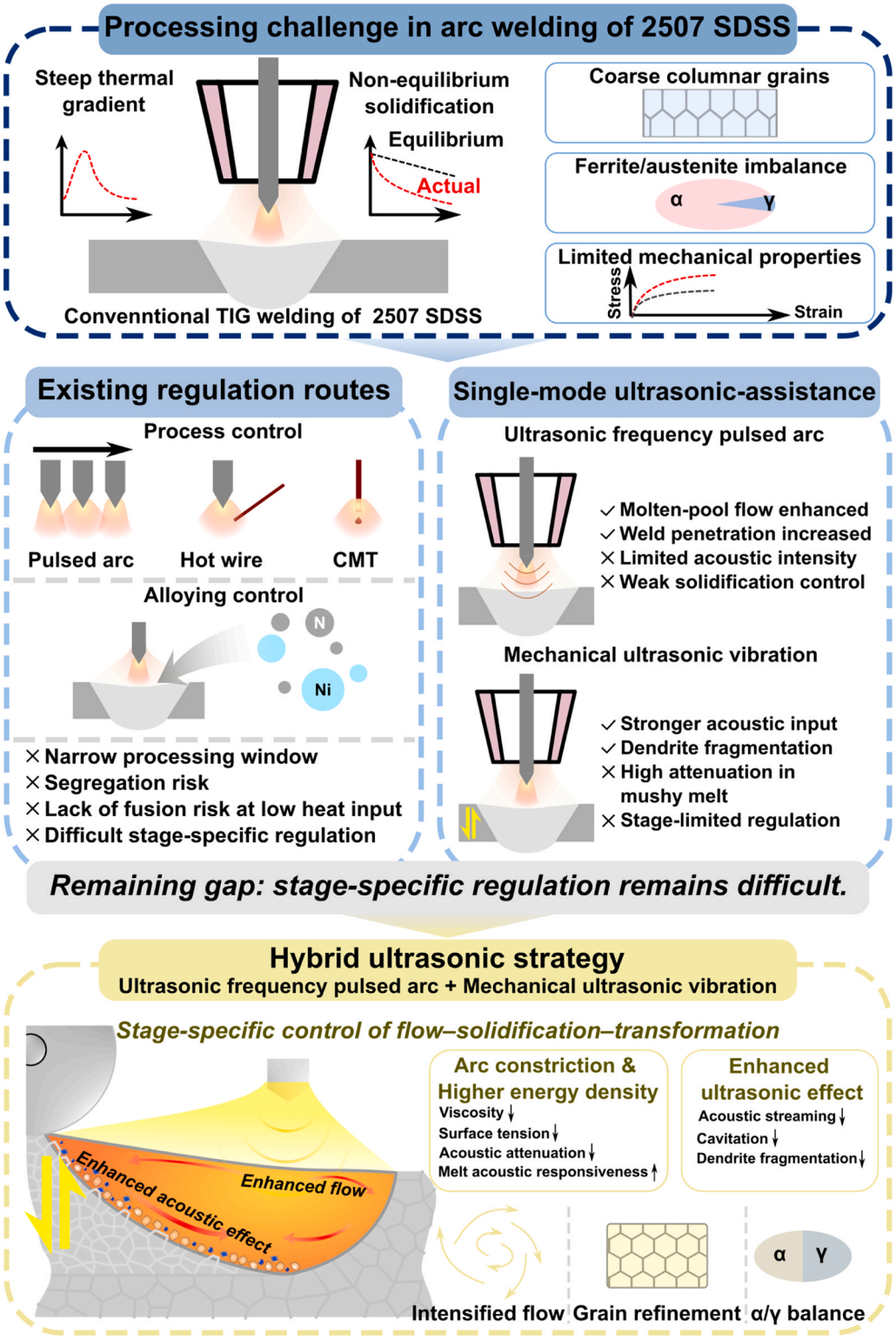


Fig. 1. Challenges in arc-based melt processing and concept of the hybrid ultrasonic strategy in this work. Arc welding and related melt-based manufacturing processes are often limited by steep thermal gradients and non-equilibrium solidification, which can lead to coarse columnar grains, phase imbalance and poor mechanical properties. Existing process-control, alloying and single-mode ultrasonic strategies provide only partial regulation and are insufficient for stage-specific control of molten-pool flow, solidification and phase transformation. In this work, an ultrasonic-frequency pulsed arc is coupled with mechanical ultrasonic vibration to form a hybrid ultrasonic strategy, aiming to exploit their functional complementarity and achieve coordinated regulation across the welding process. Schematic illustrations were drawn based on [4,7,8,19,26,27].

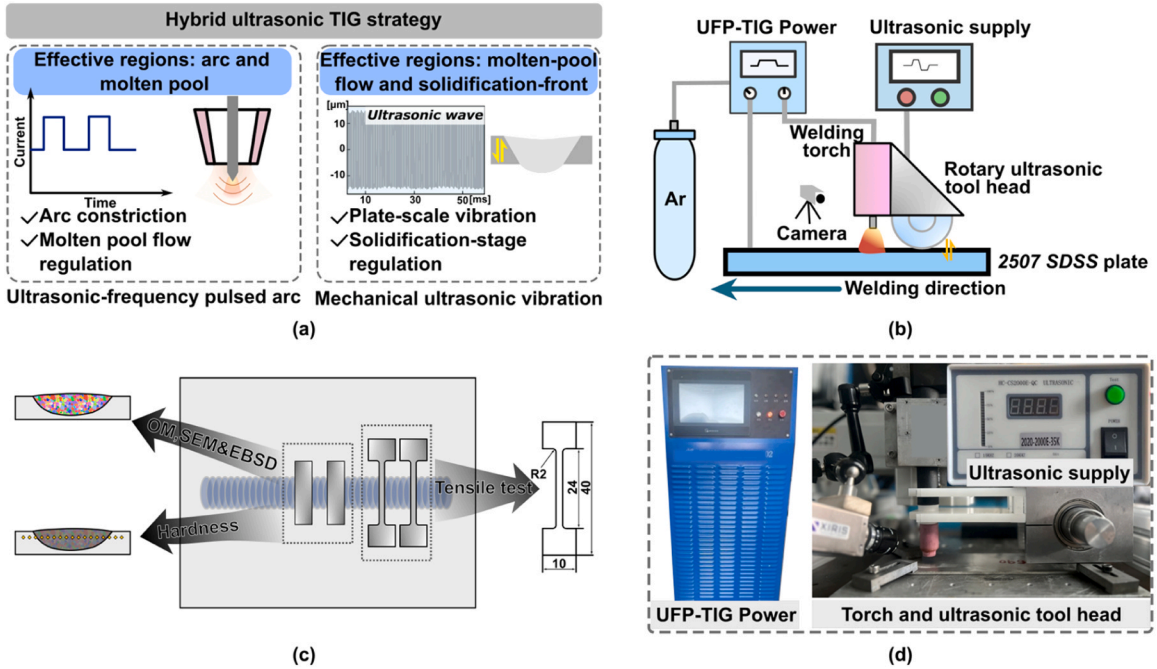


Fig. 2. Process concept, experimental apparatus and sampling design of hybrid ultrasonic TIG welding. (a) Functional partition of the two ultrasonic inputs. The ultrasonic-frequency pulsed arc mainly acts in the arc and molten-pool regions by promoting arc constriction and molten-pool flow regulation, whereas the mechanically introduced vibration mainly affects the molten-pool flow and solidification-front regions through plate-scale vibration and solidification-stage regulation. (b) Schematic diagram of the hybrid ultrasonic-assisted welding system; (c) specimen sampling locations for microstructural characterization, hardness measurement and tensile testing; (d) photograph of the welding experimental system, showing the UFP-TIG power source, welding torch, ultrasonic roller.

Table 1
The welding parameters.

Process type	Weldingspeed(mm/min)	UFP-current/I_p (A)	Background current/I_b (A)	Frequency (kHz)	Ultrasonic power(W)	Background Ratio (%)
C-TIG	200	0	180	0	0	100
UFP-TIG	200	100	130	20	0	50
UV-TIG	200	0	180	0	800	100
HU-TIG	200	100	130	20	800	50

welding torch, with the distance between the torch axis and the roller contact point along the travel direction set to 170 mm. This trailing position was selected according to previous vibration-field simulations [28], which indicated that the ultrasonic vibration amplitude in the weld region was relatively high under this spacing. Meanwhile, the trailing configuration reduced direct mechanical interference with the arc and molten-pool surface while maintaining continuous vibration input to the welded plate.

2.2. Welding procedure and processing parameters

To compare non-equilibrium solidification under different acoustic conditions, 2507 SDSS welds were fabricated using four welding conditions: conventional TIG (C-TIG), ultrasonic-frequency pulsed TIG (UFP-TIG), ultrasonic vibration-assisted TIG (UV-TIG), and hybrid ultrasonic TIG (HU-TIG). These four conditions were designed to isolate the individual and coupled effects of the two ultrasonic inputs. C-TIG was used as the baseline condition. UFP-TIG was used to evaluate the effect of ultrasonic-frequency current modulation and arc-induced excitation. UV-TIG was used to assess the effect of mechanically introduced ultrasonic vibration alone. HU-TIG was used to determine whether the two inputs could produce a coupled thermo-acoustic effect beyond either single ultrasonic mode. The base material was 2507 SDSS with a nominal composition of 25Cr–7.3Ni–4Mo–0.25 N, in dimensions of 250 mm × 150 mm × 3 mm. In normal SDSS welding, nitrogen is added to the shielding gas to stabilize austenite. However, to eliminate nitrogen-related effects and isolate the influence of ultrasonic on the weld microstructure, high-purity argon (99.9%) was used as the shielding gas. Autogenous welding was adopted to exclude the influence of filler metal. In all welding conditions, the welding speed was fixed at 200 mm/min. For UFP-TIG and HU-TIG, the ultrasonic-frequency pulsed current had a peak current of 230 A, a base current of 130 A, a duty cycle of 50%, and a frequency of 20 kHz. This peak and base current combination was selected to keep the average welding current at 180 A, identical to that used in the conventional TIG condition, as shown in Table 1. Therefore, the comparative design minimized the difference in average arc current among the four processes, while allowing the effects of current modulation and mechanical ultrasonic vibration to be distinguished.

Prior to welding, the oxide layer on the base material surface was removed by grinding, followed by drying to minimize surface contamination. During welding, the torch and the ultrasonic roller were mounted on the same three-axis motion platform and moved synchronously along the weld path. After welding, the welds were cooled to room temperature before deactivating the roller.

2.3. Process monitoring, microstructure characterization and mechanical testing

The welding process was monitored to evaluate the influence of ultrasonic-frequency pulsed current on the arc and molten pool. Arc morphology was recorded using a high-dynamic-range camera (XVC

1100, Xiris Automation Inc., Burlington, Canada). The molten-pool surface temperature was measured using an infrared thermal camera (XIR 1800, Xiris Automation Inc., Burlington, Canada). Arc radiation interferes with absolute temperature measurement, so infrared data were used to compare temperature evolution across welding conditions.

Specimens were prepared using electrical discharge wire cutting. Metallographic and tensile test samples were extracted perpendicular and parallel to the welding direction, respectively. The metallographic specimens were ground and polished, then etched with Beraha reagent for microstructural observation. For electron backscatter diffraction (EBSD) analysis, samples were polished using a planar vibratory method to remove surface stress layers. Optical microscopy (OM) was used for microstructural examination, while EBSD analysis was conducted using a scanning electron microscope (SEM, ZEISS Gemini-560) equipped with an EBSD detector. Fig. 2c shows the schematic of the specimen sampling location and size.

Vickers hardness was measured across the weld cross-section using a Vicker hardness tester (HVS-1000A, Jinan Huayin Testing Instrument, Jinan, China). Tensile tests were performed at room temperature using specimens extracted parallel to the welding direction using an electro-mechanical universal testing machine (Instron 5569, Illinois Tool Works Inc., Glenview, USA). At least three specimens were tested per condition. Fracture surfaces were examined by SEM to identify fracture features.

3. Results

3.1. Weld-zone microstructure under different ultrasonic conditions

Fig. 3 shows the weld macrostructures under different ultrasonic conditions. The application of ultrasonic-frequency pulsed current caused electromagnetic pinching and thermal constriction of the arc, reducing arc diameter and increasing energy density [30]. This heat source densification improved thermal efficiency, increasing weld penetration by 32.6% and width by 9.6% compared to C-TIG. Mechanical vibration alone achieved a similar depth (1.88 mm) but did not broaden the weld width in UV-TIG. The combination of arc-induced ultrasonic and mechanical vibration yielded a synergistic effect, resulting in 2.83 mm penetration (+100.7%) and 10.66 mm weld width (+19.9%) over C-TIG in HU-TIG. These results indicate that hybrid strategy enhances arc-to-pool heat transfer and broadens the fusion zone, suggesting improved energy utilization. Moreover, under the hybrid ultrasonic, the increase in penetration significantly exceeds the sum of depth increments achieved by either ultrasonic mode acting independently.

Fig. 4 shows the microstructure of the weld zone under different ultrasonic fields. The OM images (Fig. 4a1, b1, c1, and d1) reveal evident ultrasonic-induced grain refinement. Compared to C-TIG, both UFP-TIG and UV-TIG achieved moderate grain refinement. HU-TIG showed the most significant improvement, attributed to the combined effects of pulsed arc excitation and mechanical vibration. The longitudinal sections (Fig. 4a3, b3, c3, and d3) show that the pronounced columnar structure observed in C-TIG is gradually suppressed with the application of ultrasonic; nevertheless, grain growth remains generally aligned with the thermal gradient. In HU-TIG, columnar grains are significantly shortened, indicating superior grain disruption compared to either ultrasonic mode alone. SEM views (Fig. 4e) show ubiquitous rod-like Cr_2N precipitates both within ferrite grains and along ferrite/austenite boundaries. Because nitrogen has limited solubility in ferrite and the cooling rate is similar across all four processes, supersaturated nitrogen tends to precipitate as Cr_2N. However, ultrasonic assistance alters fluid flow rather than chemical composition and therefore cannot suppress this precipitation reaction. In addition, TEM characterization (Fig. 4f) reveals a high dislocation density in ferrite after HU-TIG welding, which is consistent with repeated acoustic shock loading

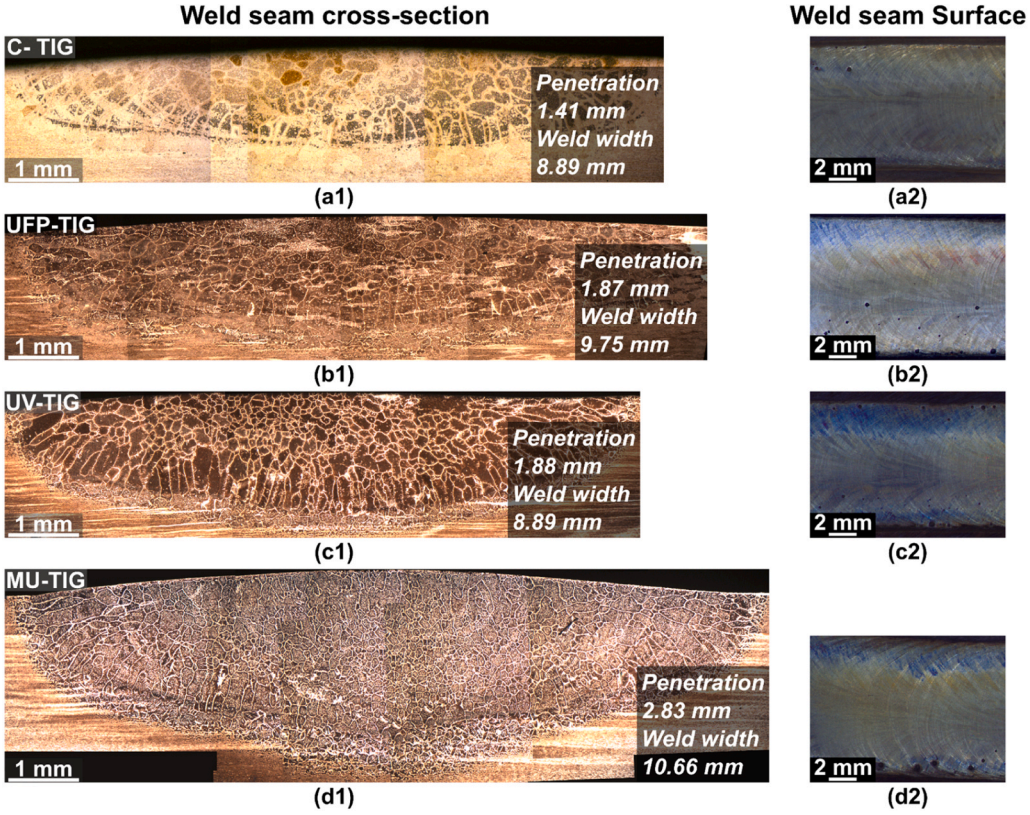


Fig. 3. Weld macrostructures under different ultrasonic conditions: etched cross-sections and weld-bead surface appearance. (a1, a2) C-TIG; (b1, b2) UFP-TIG; (c1, c2) UV-TIG; (d1, d2) HU-TIG. UFP-TIG and UV-TIG modify weld penetration in different ways, whereas HU-TIG produces the deepest and widest fusion zone, indicating the coupled effect of hybrid ultrasonic on arc-to-pool heat transfer and molten-pool flow.

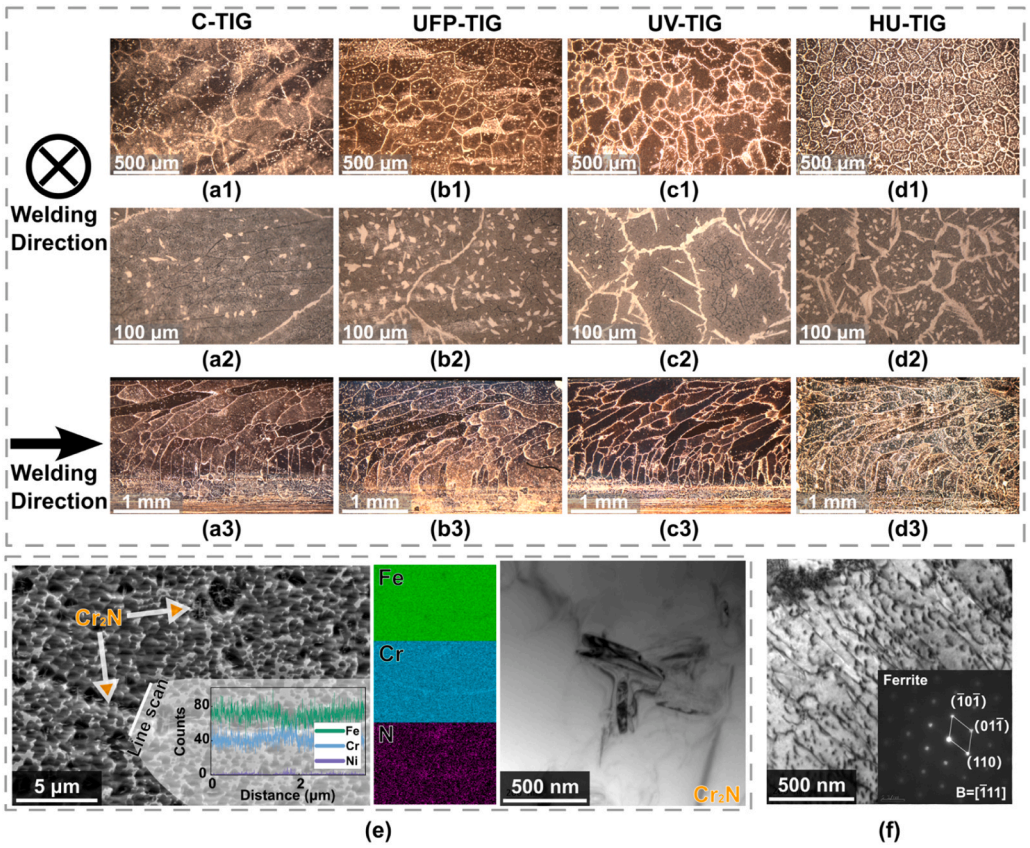


Fig. 4. Weld zone microstructures under different ultrasonic conditions. Etched optical micrographs of weld zone: (a1–a3) C-TIG; (b1–b3) UFP-TIG; (c1–c3) UV-TIG; (d1–d3) HU-TIG. The welds show different degrees of columnar-grain suppression and microstructural refinement, with HU-TIG exhibiting the most refined weld-zone structure. (e) SEM and EDS observations reveal Cr_2N precipitates and the corresponding Fe, Cr and N distributions in the weld zone. (f) TEM observation of HU-TIG shows a high dislocation density in ferrite, suggesting that hybrid ultrasonic assistance may introduce additional defect structures during solidification.

during solidification. These results suggest that hybrid ultrasonic not only refine grain morphology but also elevate dislocation density, offering additional nucleation sites for phase transformation and enhanced mechanical performance.

3.2. *Texture evolution and ferrite–austenite transformation in the weld zone*

Fig. 5 shows the grain morphology and orientation in the weld zone under different ultrasonic conditions. In C-TIG (Fig. 5a), coarse ferrite grains show a dominant [001] orientation, while austenite is sparse and randomly oriented. The average ferrite grain size reaches 143.4 μm, whereas austenite grains remain very fine (3.0 μm). This aligns with the typical solidification path of SDSS [31], where ferrite crystallizes first from the liquid, followed by austenite formation along grain boundaries due to solute redistribution. Additionally, a portion of austenite is formed through solid-state transformation within the ferrite matrix [32]. However, under single-pass welding, the thermal driving force is limited, resulting in minimal intragranular austenite with small size.

In UFP-TIG (Fig. 5b), ferrite grains predominantly align along [011], while boundary austenite aligns with [111]. While ferrite grain size remains similar to that of C-TIG, the austenite morphology changes markedly. Austenite in UFP-TIG exhibits typical Widmanstätten morphology, forming needle-like or plate-like structures along the {111} planes of ferrite. These structures disrupt matrix continuity and induce plastic strain incompatibility. The increased α/γ interface area may promote microgalvanic coupling, which is generally regarded as aggravating corrosion susceptibility [33,34]. In UV-TIG (Fig. 5c), the average ferrite grain size decreases to 90.8 μm. While some [111]-oriented grains appear, the overall orientation intensity is relatively weak (max intensity: 3.8). HU-TIG (Fig. 5d) yields the finest microstructure, reducing ferrite grain size to 70.2 μm—51.0%, 52.4%, and 22.6% smaller than that of C-TIG, UFP-TIG, and UV-TIG, respectively. The microstructural refinement induced by the hybrid ultrasonic strategy is not merely the arithmetic sum of the effects achieved by each individual ultrasonic mode but rather exhibits a pronounced synergistic refinement. Moreover, both ferrite and austenite grains exhibit randomized orientations, indicating weakened crystallographic texture under hybrid ultrasonic excitation.

The orientation relationship and distribution of ferrite and austenite under different ultrasonic conditions were further described in Fig. 6. In SDSS welds, ferrite-austenite interfaces that satisfy the Kurdjumov-Sachs (K-S) orientation relationship are mechanically unfavourable, because K-S boundaries require coherent shuffling of all atoms across the interface, they accommodate plastic strain poorly and act as sites of stress concentration and crack initiation [35,36]. To quantify this effect, boundaries with misorientations $\leq 6^\circ$ from any of the 24 ideal K-S variants were extracted (white lines in Fig. 6a-d). Three morphological types of K-S interfaces were observed. Grain-boundary austenite (Fig. 6f1) shows K–S alignment with adjacent ferrite, while the opposite side appears randomly oriented. Widmanstätten austenite fully complies with the K-S relationship with respect to ferrite. In contrast, intragranular austenite, appearing as fine islands within ferrite, displays random orientation (Fig. 6f2, f3).

Quantitative results (Fig. 6e) reveal changes in K-S boundary fractions and phase ratios under different ultrasonic conditions. In C-TIG, K-S interfaces account for 47.0% of the total ferrite–austenite boundaries. In UFP-TIG, although K–S boundary length increases by 25% due to Widmanstätten lath formation, the relative fraction remains 43.7% due to the expanded total boundary length. UV-TIG refines ferrite grains and

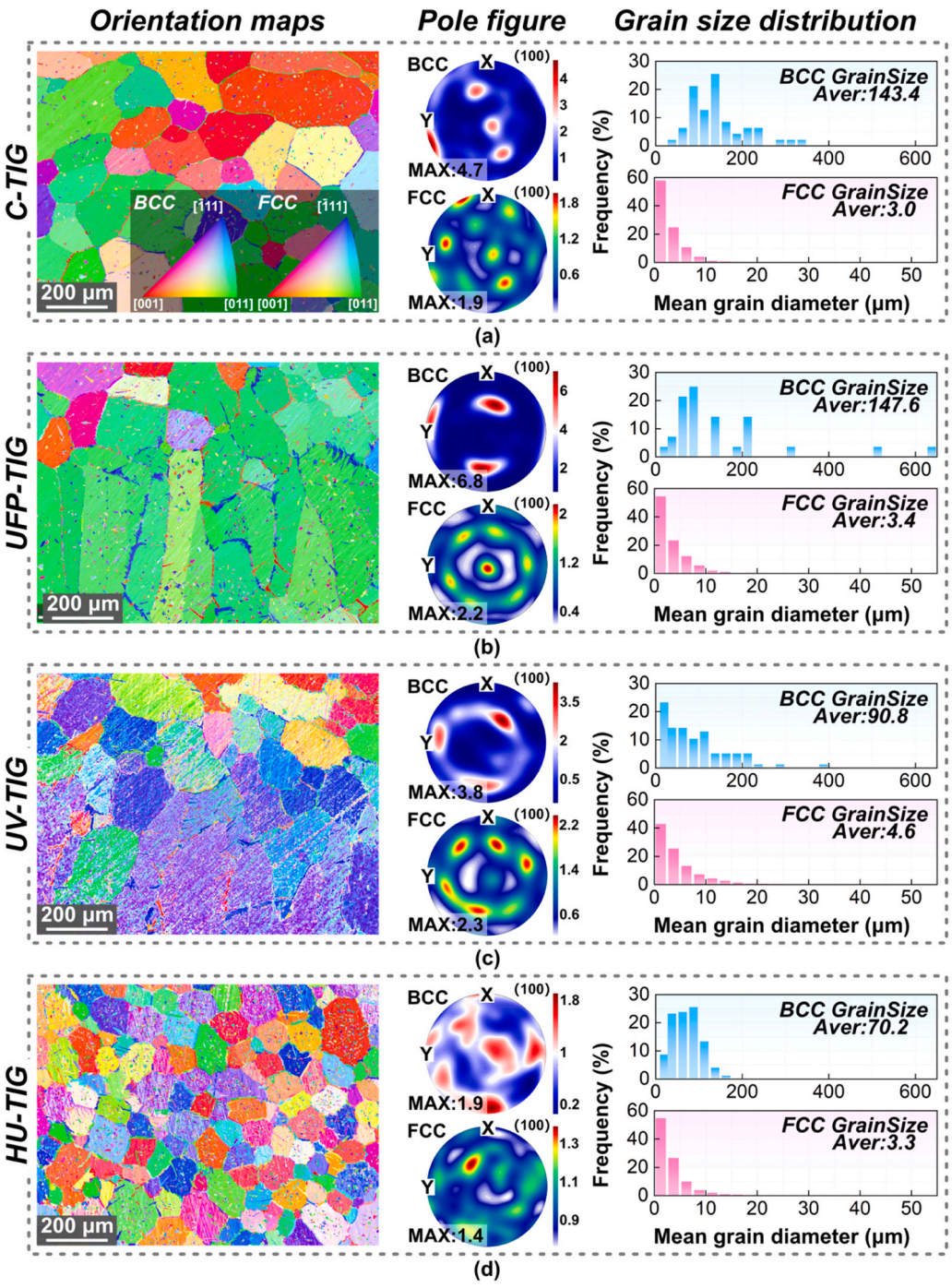


Fig. 5. EBSD results of weld microstructure under different welding methods. Orientation maps, pole figure, and grain size distribution: (a) C-TIG; (b) UFP-TIG; (c) UV-TIG; (d) HU-TIG. C-TIG shows coarse ferrite grains and a relatively strong solidification texture, whereas HU-TIG exhibits the finest ferrite grains and a weakened orientation concentration. These results indicate that hybrid ultrasonic assistance promotes grain refinement and reduces preferred crystallographic growth more effectively than single ultrasonic mode.

increases total interface length, reducing the K–S boundary fraction to 38.6%. HU-TIG, with the finest grains and densest boundaries, further reduces the K–S fraction to 20.3%. Meanwhile, the austenite fraction triples—from 6.9% in C-TIG to 20.0% in HU-TIG—highlighting a 190% relative increase. These results indicate that hybrid ultrasonic assistance not only refines the microstructure but also improves the uniformity of austenite distribution.

3.3. Mechanical properties and fracture morphology under different ultrasonic conditions

Fig. 7 shows the mechanical properties and tensile fracture morphology. The refined, uniform microstructure in HU-TIG yields substantial mechanical improvements over other methods. Specifically, the tensile strength of HU-TIG increases by 16.22%, 5.60%, and 3.34% compared to C-TIG, UFP-TIG, and UV-TIG, respectively. Elongation improves even more markedly—by 97.4% over C-TIG and 20.2% over UV-TIG. Analogously to grain refinement, the elongation increment

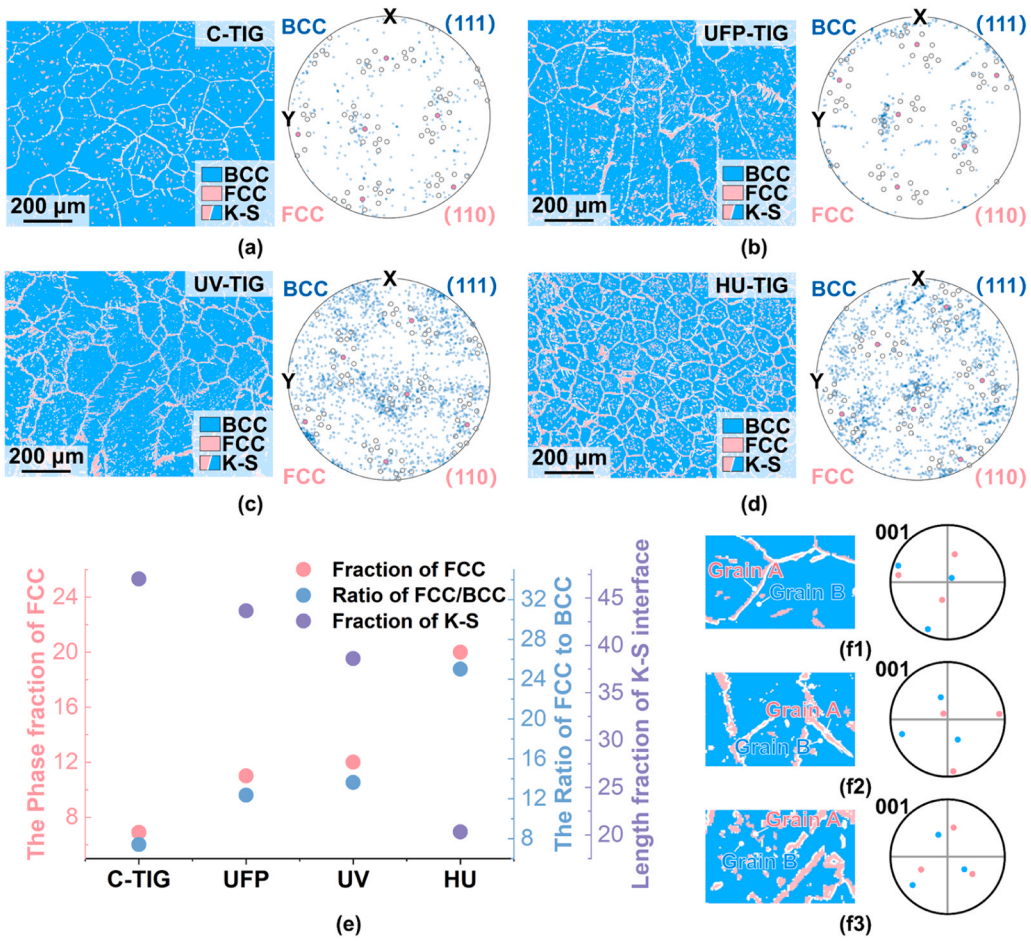


Fig. 6. Ferrite–austenite phase distribution and K–S interface analysis. (a-d) Phase maps of C-TIG, UFP-TIG, UV-TIG, and HU-TIG; (e) quantitative comparison of austenite fraction and K–S interface fraction under different ultrasonic conditions.; (f1-f3) representative morphologies of K-S interfaces. HU-TIG increases the austenite fraction and markedly reduces the fraction of K–S interfaces.

achieved under the hybrid strategy exceeds the arithmetic sum of those attained individually by UFP-TIG and UV-TIG. Such enhancements in both tensile strength and ductility are primarily attributed to the refined and uniformly distributed microstructure enabled by the hybrid ultrasonic. The fracture morphology further confirms the benefits of microstructural refinement. For the C-TIG welds (Fig. 7d1, d2), the fracture surface contains relatively planar regions with pronounced secondary cracking. At higher magnification, a long open crack is observed together with shallow dimples around the crack flanks. This indicates a mixed fracture mode in which localized intergranular cracking is superimposed on micro void coalescence. In contrast, the fracture surface of HU-TIG (Fig. 7g1, g2) shows a much more uniform and ductile fracture pattern, with evenly distributed ductile dimples, reflecting the finer, more uniform grain structure that improves the overall toughness and ductility of the weld.

4. Discussion

4.1. Effect of hybrid ultrasonic assistance on molten-pool dynamics and weld morphology

As shown in Fig. 3, the morphology of the weld cross-section varies significantly under different ultrasonic conditions. Compared with C-TIG, both UFP-TIG and UV-TIG enlarge the molten pool, but in different ways: mechanical vibration primarily increases penetration, whereas arc-induced ultrasonic simultaneously enhances both penetration and width. The hybrid ultrasonic achieves the most pronounced effect, with substantial increases in both dimensions. The penetration and morphology of the molten pool are primarily regulated by heat input and the molten pool convection. Although the introduction of hybrid ultrasonic leads to a slight increase in the total heat input, this increment is basically equal to the linear superposition of the heat input increments caused by each of the two single ultrasonic modes, and the absolute value is relatively small. However, the geometry of HU-TIG indicates that the expected increase in the penetration due to the change in heat input alone is far from sufficient to explain the significant morphological evolution of molten pool. Therefore, it is reasonable to infer that there is a synergistic coupling effect between the ultrasonic-frequency pulsed arc and the mechanical ultrasonic vibration. To clarify the underlying mechanisms, the effects of each ultrasonic field are analyzed individually before considering their synergy.

As illustrated in Fig. 8a, the high-frequency pulsed current constricts the arc plasma due to electromagnetic pinching and thermal delay effect, increasing the proportion of high-energy regions and improving overall heat transfer efficiency [29]. Infrared thermography measurements (Fig. 8b) reveal that the molten pool surface temperature rises by ~550 °C compared to C-TIG, confirming enhanced arc heating. Although absolute temperature values are influenced by arc interference, the results qualitatively demonstrate the intensified heating effect. Elevated pool temperature lowers surface tension in the center of molten pool, thereby strengthening Marangoni convection. The intensified inward flow promotes deeper penetration and wider fusion zones, consistent with other observations [37,38].

In UV-TIG, mechanical vibration introduces forced vibrations into

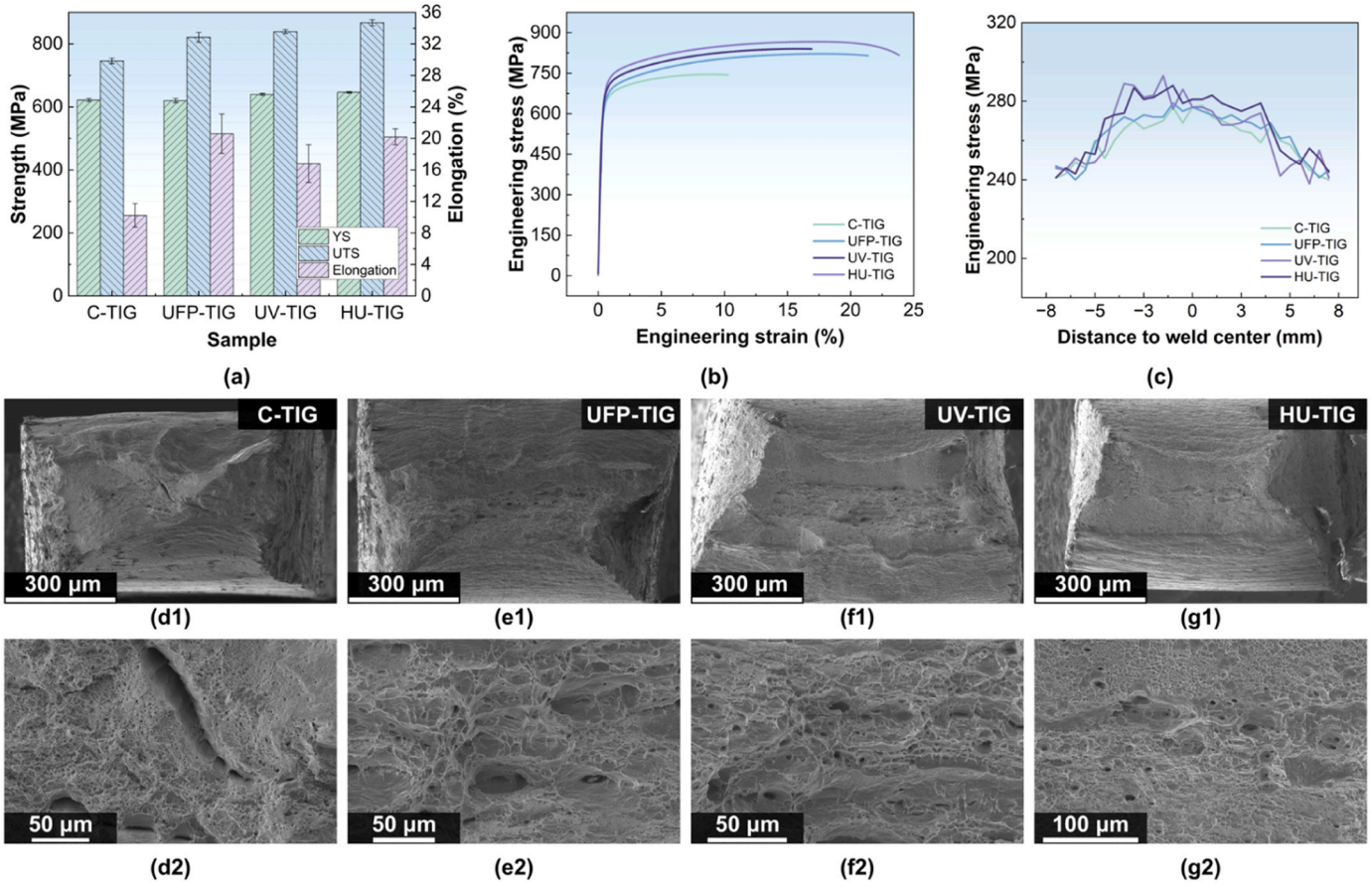


Fig. 7. Tensile properties, hardness distribution and fracture morphology under different ultrasonic conditions. (a) Tensile properties, (b) tensile engineering stress-stain curve and (c) Vickers hardness of different welds; tensile-fracture surfaces: (d1, d2) C-TIG; (e1, e2) UFP-TIG; (f1, f2) UV-TIG; (g1, g2) HU-TIG. HU-TIG exhibits a more ductile fracture morphology with more uniformly distributed dimples, which is consistent with the improved tensile strength and elongation. (Error bars represent ± STDEV from value (n = 3)).

the base metal. When the base metal is heated by the arc, the accelerated atomic motion at the solid–liquid interface facilitates melting of the underlying metal, resulting in increased penetration. According to Peng et al., the morphology on both sides of the weld cross-section is not significantly related to heat conduction but is mainly determined by the Marangoni force [39]. However, since ultrasonic energy is significantly dissipated within the mushy zone, mechanical vibration scarcely alters Marangoni flow. Consequently, the weld width and morphology remain nearly unchanged.

In HU-TIG, the hybrid action of arc-induced ultrasonic and mechanical vibration concentrates the heat source and enhances thermal conduction efficiency. The elevated temperature in the molten pool reduces melt viscosity, as shown in Fig. 8c, which promotes stronger Marangoni convection and accelerates mass transport toward the pool root. At the same time, the acoustic attenuation of the melts decreases with rising temperature, enabling the molten pool to respond more effectively to ultrasonic excitation. This facilitates acoustic streaming, which further intensifies the "pool-digging" effect and increases penetration [23]. The combined thermal and acoustic effects thus reshape the molten pool into the significantly deeper and wider morphology observed in Fig. 3, as schematically illustrated in Fig. 8d. The hybrid ultrasonic strategy not only enhances energy utilization efficiency but also establishes a favorable fluid dynamic environment, thereby reshaping the molten pool morphology more effectively than the use of any single ultrasonic mode alone. This also endows this strategy with the potential for expansion to other metal processing techniques that require precise control of the molten pool morphology.

4.2. Thermo-acoustic coupling, acoustic pressure enhancement and cavitation tendency

In HU-TIG, the initial intention was for arc-induced and mechanical ultrasonics to act at different stages. However, the process monitoring data and the results of microstructure characterization jointly indicate that the regulatory effect of hybrid ultrasonic strategy on molten pool flow behavior and the solidification process is not a simple superposition of the two individual ultrasonic effects but rather exhibits a significant enhancement. Based on the characteristics of the two ultrasonic modes, we speculate that the cooperative mechanism may mainly be reflected in the following two aspects: temperature-induced reduction in acoustic attenuation and wave superposition.

The ultrasonic-frequency pulsed arc provides efficient heating, significantly elevating the molten pool temperature. This temperature rise profoundly alters the thermophysical properties of the melt, most notably causing a sharp decrease in its viscosity (η), as shown in Fig. 8c. Acoustic attenuation in a liquid medium is primarily governed by viscous and thermal relaxation mechanisms. The contribution from viscous attenuation, α_v can be qualitatively described by the classical relationship [40,41]:

$$\alpha_v \propto \frac{\eta}{\rho c^3} f^2 \tag{1}$$

where ρ is the density, c is the sound velocity, η the liquid viscosity, and f is the frequency. As shown in Fig. 8c, viscosity decreases much more sharply with temperature than either density or sound velocity. Using the measured molten pool temperatures and temperature-dependent

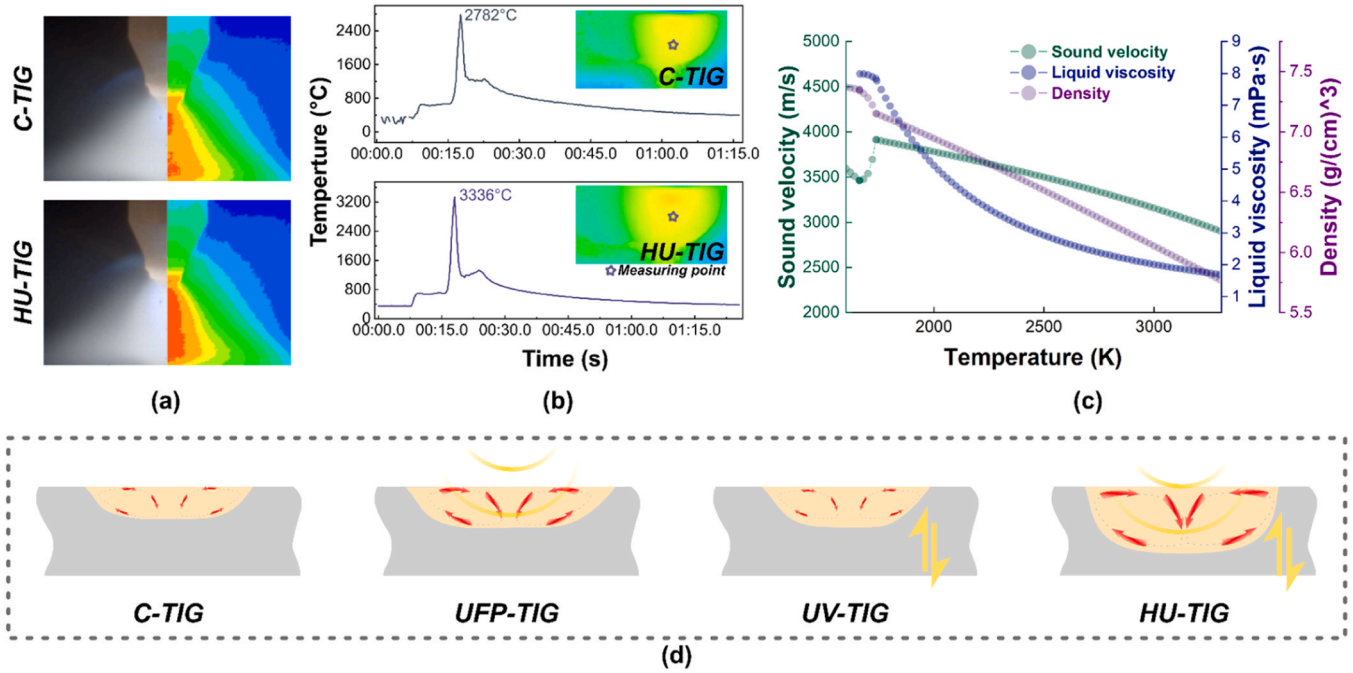


Fig. 8. Arc morphology, molten-pool temperature, melt thermophysical properties, and molten pool flow evolution schematic. (a) Arc morphology captured by high dynamic range camera; (b) temporal evolution of molten pool surface temperature measured by infrared thermography, with the marked region indicating the monitoring position; (c) temperature-dependent thermophysical properties of 2507 stainless steel; (d) schematic evolution of molten-pool morphology and flow behavior under C-TIG, UFP-TIG, UV-TIG and HU-TIG. The results indicate that ultrasonic-frequency pulsed arc increase molten-pool temperature and modify the melt thermophysical state, thereby strengthening Marangoni convection, acoustic response and molten-pool flow regulation.

viscosity data of 2507 SDSS evaluated from thermophysical property calculations, the melt viscosity decreases of about 28.6% in HU-TIG relative to C-TIG. Therefore, reduced viscosity tends to weaken viscous dissipation and favors acoustic energy transmission.

Simultaneously, the thermal relaxation attenuation (α_T) is also reduced at elevated temperatures. This mechanism arises from the irreversible heat conduction between the compressive and rarefactive phases of the acoustic wave. Its strength depends on the ratio of the ultrasonic angular frequency (ω) to the thermal relaxation time (τ) [42, 43]:

$$\alpha_T \propto \frac{\omega^2\tau}{1 + (\omega\tau)^2} \tag{2}$$

The τ characterizing the rate at of local temperature equilibration, decreases sharply with increasing temperature due to enhanced atomic mobility. The reduction in τ moves the system away from the condition of peak thermal attenuation, thereby further reducing the total attenuation coefficient ($\alpha_v + \alpha_T$). The decreased overall attenuation enables deeper penetration of acoustic energy, significantly improving the acoustic responsiveness of the molten pool.

Chen's finite element modelling was employed to evaluate the acoustic pressure distribution within the molten pool in HU-TIG [44]. Fig. 9a shows the simplified multi-physics model integrating both acoustic sources. Fig. 9b compares the simulated acoustic pressure transients near the bottom of the molten pool. The simulated peak acoustic pressure amplitude near the pool bottom increases from 0.52 MPa in UV-TIG to 0.84 MPa in HU-TIG. In UFP-TIG, the arc-induced acoustic pressure is weak and attenuates rapidly. In UV-TIG, mechanical vibration transmitted through the substrate generates a higher acoustic pressure. The simulated peak acoustic pressure in HU-TIG exceeds the arithmetic sum of the individual peak pressures under UFP-TIG and UV-TIG, providing evidence that the acoustic field superposition in hybrid ultrasonic coupling is nonlinear rather than linearly additive. Notably, in HU-TIG, the simulated acoustic pressure amplitude increases substantially due primarily to the markedly reduced

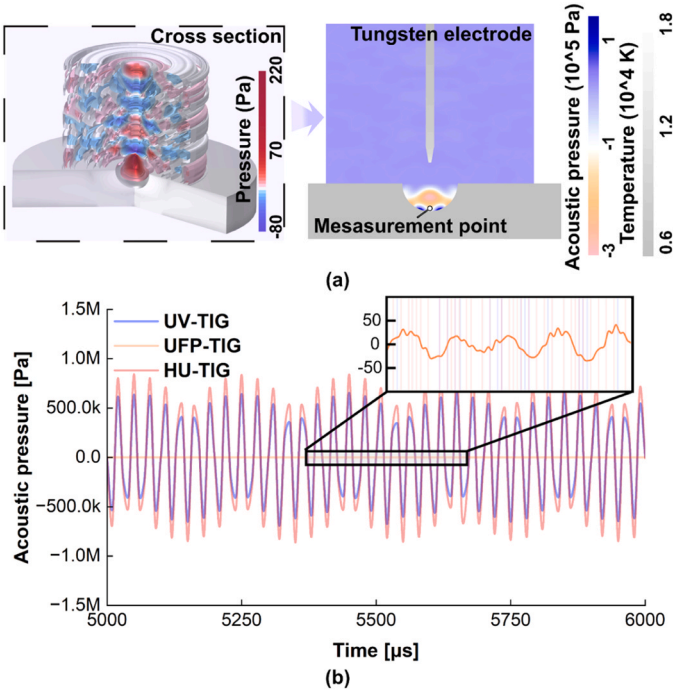


Fig. 9. Idealized acoustic pressure simulation and cavitation tendency in the hybrid ultrasonic process. (a) Idealized acoustic model used to evaluate the coupled action of ultrasonic-frequency pulsed arc excitation and mechanical ultrasonic vibration, with the measurement point located near the molten-pool bottom; (b) simulated acoustic pressure histories at the selected point for UFP-TIG, UV-TIG and HU-TIG. HU-TIG exhibits a higher acoustic pressure amplitude than either single ultrasonic mode, indicating enhanced acoustic-energy transmission in the molten pool of HU-TIG. The increased acoustic pressure, together with the reduced surface tension at elevated temperature, suggests a higher cavitation tendency under hybrid ultrasonic.

acoustic attenuation in the hotter pool. Although wave interference occurs, its contribution is secondary compared to this dominant thermophysical effect.

Cavitation requires the acoustic pressure during the rarefaction phase to exceed a critical threshold, expressed by [45]:

$$p_c \approx p_0 - p_v + \sqrt{\frac{2\sigma}{R_0}} \tag{3}$$

where p_0 is the hydrostatic pressure, p_v is the vapor pressure, σ is the surface tension, and R_0 is the initial bubble radius (about 10 μm in metal cavitation). And the p_v is significantly smaller than p_0, rendering it typically negligible. The σ decreases observably with temperature, as described [46]:

$$\sigma(T) = \sigma_m - k(T - T_m) \tag{4}$$

Where σ_m is the surface tension at melting point (1.82 $N{\cdot}m^{-1}$ for 2507 SDSS), T_m is the critical temperature which is often taken as the melting point, and k is temperature coefficient of surface tension ($4.2{\cdot}10^{-4}$ $N{\cdot}m^{-1}{\cdot}K^{-1}$, experimentally fitted for steel melt [47]. As temperature increases, σ decreases, the p_c (C-TIG) has been estimated as 0.599 MPa, and the value of p_c (HU-TIG) is 0.548 MPa.

Therefore, in the HU-TIG process, two effects synergistically enhance cavitation: the increased acoustic pressure amplitude due to reduced attenuation, and the lowered cavitation threshold due to decreased surface tension. This combined effect markedly enhances potential acoustic cavitation activity throughout the molten pool.

4.3. Grain refinement by solidification-condition regulation and acoustic effects

The refinement of weld microstructure under hybrid ultrasonic assistance is evident, as shown in Fig. 4, where the average grain size in the weld zone is reduced by nearly 50% compared with C-TIG. Although ultrasonic introduces additional energy into the molten pool, a thermodynamic factor potentially conducive to grain coarsening, the consistent and pronounced grain refinement observed in both UV-TIG and HU-TIG processes confirms that the dynamic effects of ultrasonic decisively outweigh its thermal contribution, resulting in a net refinement of the solidification microstructure. Crucially, the grain size reduction in HU-TIG exceeds that achieved with UV-TIG, which further corroborate there is a mutually reinforcing, non-additive interplay between the two ultrasonic modalities. In conventional welding of 2507 SDSS, the grain morphology is primarily governed by the thermal gradient (G) and the solidification growth rate (R). To clarify whether the observed grain refinement is primarily attributed to alterations in solidification conditions or to acoustic-induced effects, the contributions of each factor were decoupled and analyzed within the HU-TIG process.

The evolution of thermal conditions was firstly analyzed in HU-TIG. The dendrite tip growth velocity (V) in arc welding can be approximated as [48,49]:

$$V = V_L \cos\theta \tag{5}$$

Where V_L is the welding speed and θ is the angle between the grain growth direction and welding direction, as shown in Fig. 10a. Based on this relation, dendrite growth velocities of V_C= 0.0029 m/s for C-TIG and V_{HU}= 0.0032 m/s for HU-TIG were obtained. Infrared

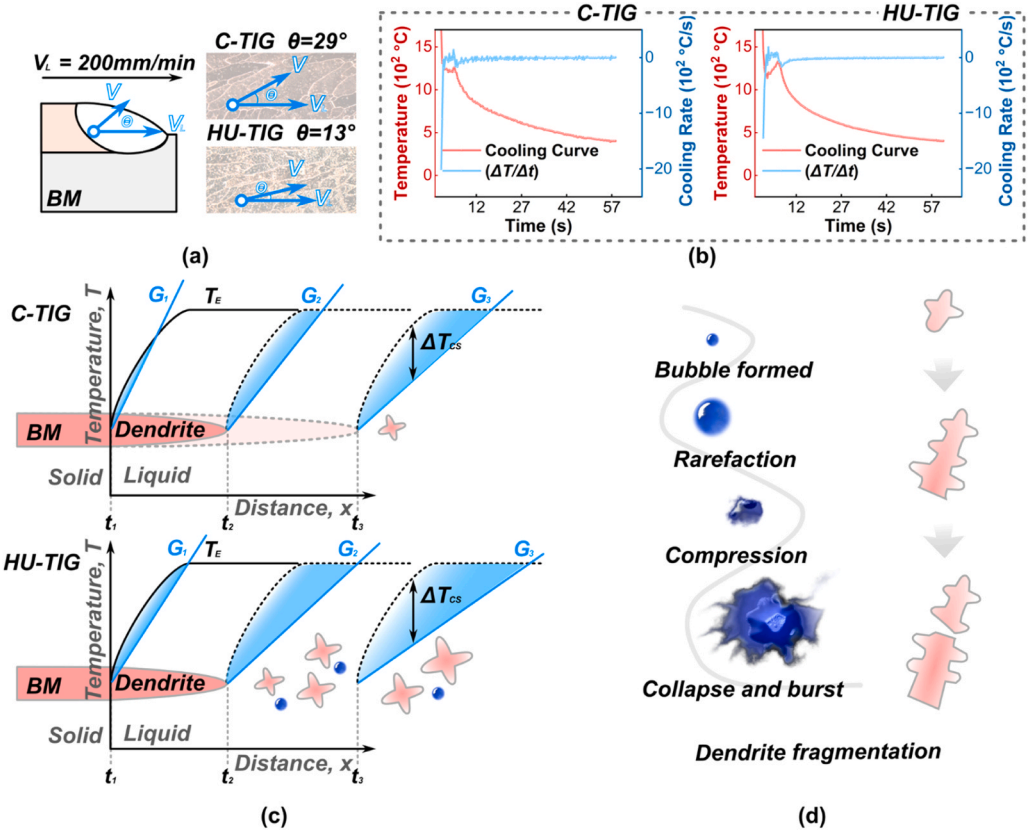


Fig. 10. Solidification behavior and grain refinement mechanism under the hybrid ultrasonic strategy. (a) Schematic definition of grain growth direction relative to the welding travel direction in the longitudinal section of the weld pool; (b) cooling curves and cooling rates during solidification for C-TIG and HU-TIG; (c) schematics of comparison of the constitutional-supercooling region and solid–liquid interface growth morphology under C-TIG and HU-TIG; (d) schematic of potential cavitation-bubble evolution and dendrite fragmentation under alternating acoustic pressure. The comparison indicates that grain refinement in HU-TIG is associated with the combined effect of reduced thermal gradient, enlarged constitutional-supercooling region and acoustic assisted dendrite fragmentation.

thermography was used to determine cooling rates near the solidus, yielding G of C-TIG and HU-TIG were calculated G_C is 242℃/mm, and G_{HU} is 171.9℃/mm, as shown in Fig. 10c. The reduced G under hybrid ultrasonic increases constitutional undercooling at the solid–liquid front, which favors heterogeneous nucleation ahead of dendrite tips (Fig. 10c).

According to the extended Hunt [50,51], the tendency for columnar-to-equiaxed transition (CET) can be expressed in terms of the parameter $\frac{G^2}{V}$:

$$\frac{G^2}{V} = a\left[\frac{1}{3}\left(\frac{-4}{3}\frac{\pi N_0}{\ln(1-\phi)}\right)^{\frac{1}{3}}\right]^2 \tag{6}$$

Smaller $\frac{G^2}{V}$ values correspond to a stronger CET tendency, with volume fraction of equiaxed grains (ϕ)< 0.0066 representing fully columnar growth and ϕ> 0.49 representing fully equiaxed grains [52,53]. In our study, $\frac{G^2}{V}$ for HU-TIG is significantly smaller than that of C-TIG, suggesting an enhanced CET tendency. However, a similar reduction of $\frac{G^2}{V}$ was observed in UFP-TIG, where grain refinement was much less pronounced. This indicates that the CET tendency alone cannot fully account for the substantial refinement observed in HU-TIG, and that acoustic effects must play a critical role.

Unfortunately, cavitation has not yet been directly observed inside the molten pool in this study; physically, it is possible for the phenomenon to occur when the local rarefaction pressure exceeds the cavitation threshold. Cavitation bubbles nucleate and expand during the rarefaction phase and subsequently contract during the compression phase. Under the repeated high-frequency action of hybrid ultrasonic excitation, the collapse of cavitation bubbles near the solidification front generates intense impulsive stresses. The yield stress of SDSS 2507 at the liquidus temperature (1662 K) was estimated from the temperature-dependent elastic modulus relationship [20,54]:

$$\sigma_y(T) = 10^{-3}E_{300}(1+B(\frac{T-300}{T_l})) \tag{7}$$

$\sigma_y(T)$ is the yield stress at temperature T, E_{300} is the material elastic modulus at temperature 300 K, B is a coefficient, and T_l is the liquidus temperature for the material. The values for E_{300}, B and T_l are 170.1 GPa, −1 and 1662 K, respectively, of SDSS 2507. The $\sigma_y(T)$ is approximately 30.5 MPa. In contrast, even conservative estimates based on Cavitation bubble dynamics predict local shock pressures exceeding 100 MPa during bubble collapse [55,56]. Such transient stresses are several times greater than the dendrite strength, making dendrite arm fragmentation inevitable. The resulting fragments act as additional nucleation sites, further promoting equiaxed grain formation.

Therefore, the grain refinement mechanism under hybrid ultrasonic excitation is primarily governed by two key factors: the reduction of G and the enhancement of the CS zone, which promotes the CET; and cavitation-induced dendrite fragmentation, which is regarded as the dominant factor that inhibits grain growth and provides numerous nucleation sites. This hybrid strategy offers a broadly applicable scientific approach for precisely controlling microstructure via synergistic thermal regulation and acoustic field intervention in metal welding and melt-based additive manufacturing.

4.4. Effect of hybrid ultrasonic on phase transformation and interphase boundary

Hybrid ultrasonic markedly reshapes the δ-γ transformation pathway and interfacial characteristics. In HU-TIG, the γ fraction increases significantly, while the proportion of α/γ interfaces satisfying the K–S orientation relationship decreases. Conventionally, nitrogen addition in shielding gas is adopted to promote austenite precipitation; however, in this study nitrogen effects were excluded to isolate the role of ultrasonic. Consequently, the overall γ fraction in this work is lower than that in conventional welding [15,20].

The δ-γ transformation in SDSS2507 typically follows the ferrite–austenite solidification mode, with γ precipitating either during solidification as primary austenite or post-solidification as secondary austenite. The precipitating of γ can be simplified as the combined effect of thermal and acoustic factors in this work, as shown in the following equation:

$$f(\gamma)_{Total} = f(\gamma)_{Thermal} + f(\gamma)_{Acoustic} \tag{8}$$

Steep thermal gradients and retained heat promote grain boundary ferrite-to-austenite transformation, producing Widmanstätten γ plates aligned with the K–S orientation relationship in UFP-TIG. Such interfaces hinder strain accommodation and are detrimental to toughness. The $t_{12/8}$ denotes the cooling time from 1200°C to 800°C. The shorter $t_{12/8}$ promotes ferrite precipitation and longer $t_{12/8}$ promotes austenite precipitation [57]. However, the cooling time $t_{12/8}$ decreases from 8.79 s in C-TIG to 7.07 s in UFP-TIG and HU-TIG, which generally expected to inhibit γ-phase formation. In fact, the γ-phase fraction increases significantly from 6.9% to 20.0%, while the proportion of K–S boundaries drop sharply to 20.3%. This unexpected result suggests the dominant role of ultrasonic effects beyond simple thermal control in phase translation, as shown in Fig. 11.

Mechanistically, the hybrid ultrasonic field reduces liquid viscosity and enhances Marangoni convection and acoustic streaming, thinning the solute boundary layer at the δ/liquid interface. This weakens the segregation of γ-stabilizing solutes (Ni, N) in grain boundaries, facilitating intragranular γ nucleation and growth. As a result, the orientation-selective growth of Widmanstätten plates is suppressed, reducing the prevalence of K–S boundaries and yielding a more uniform γ distribution. Simultaneously, cavitation bubble collapse shears nascent γ laths and redistributes solute.

In addition, post-solidification defect structures provide heterogeneous nucleation advantages. TEM observations (Fig. 4f) reveal increased dislocation density in ferrite under HU-TIG, consistent with repeated acoustic shock and cavitation loading during solidification. These high-density defects, together with refined sub-grain boundaries, offer nucleation sites for intragranular γ precipitation. Enhanced fluid–solute transport and cavitation-assisted defect generation both promote intragranular γ nucleation, while simultaneously suppressing K–S oriented boundary growth.

4.5. Overall mechanism of the hybrid ultrasonic strategy

The key processing mechanism of the hybrid ultrasonic strategy can be understood from two aspects, as shown in Fig. 12. First, the two ultrasonic modes have different effective stages in welding. The

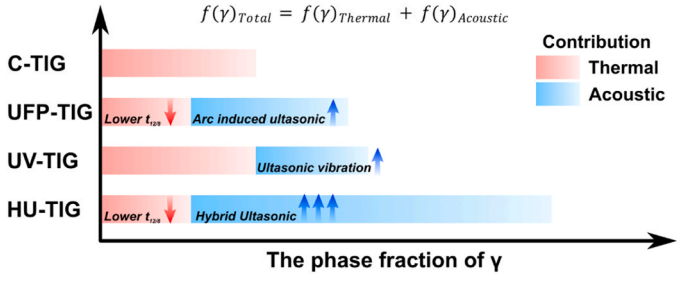


Fig. 11. Schematic illustration of thermal and acoustic contributions to γ phase precipitation in 2507 SDSS welds. The γ phase fraction is interpreted as the combined result of thermal conditions and ultrasonic-induced acoustic effects. Although the reduced cooling time in UFP-TIG and HU-TIG is expected to weaken the thermal contribution to γ formation, acoustic effects can promote solute redistribution, defect-assisted nucleation and intragranular γ precipitation.

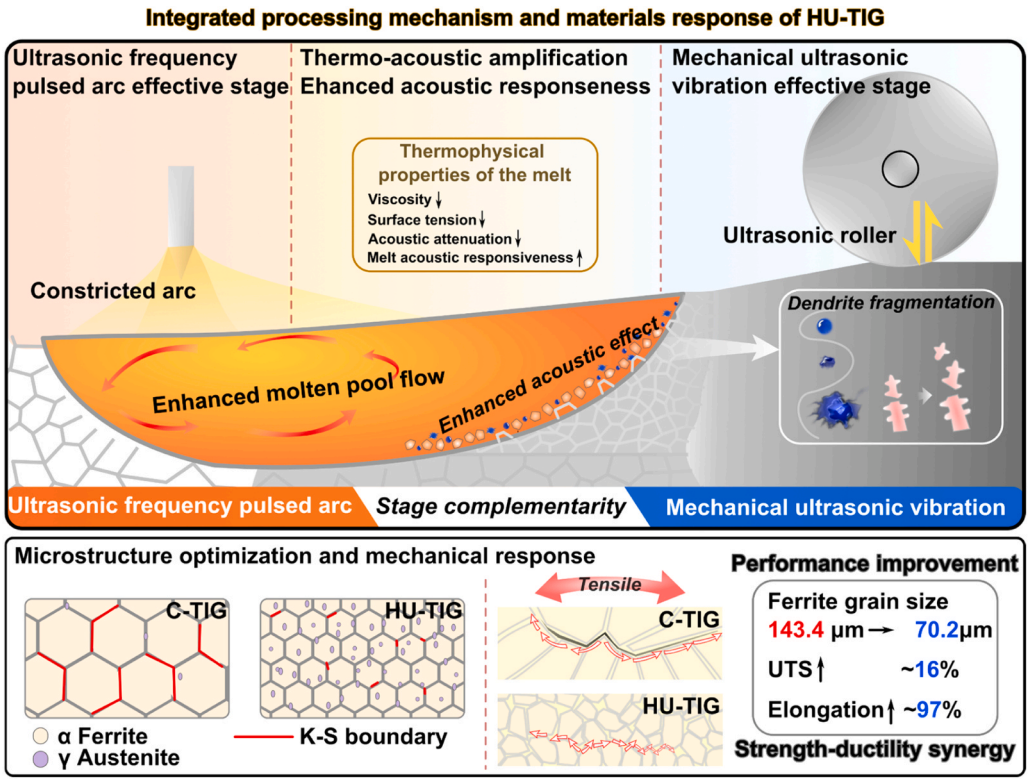


Fig. 12. Overall mechanism of the hybrid ultrasonic strategy in HU-TIG welding. The schematic illustrates the stage-complementary regulation achieved by coupling an ultrasonic-frequency pulsed arc with mechanical ultrasonic vibration. The pulsed arc mainly modifies the arc and molten pool flow stage, while the mechanical vibration strengthens acoustic loading near the solidification front. Thermal tuning of the molten pool reduces viscosity, surface tension and acoustic attenuation, improving melt acoustic responsiveness and amplifying the mechanical ultrasonic effect. The resulting thermo-acoustic regulation promotes grain refinement, phase balance and enhanced strength–ductility synergy.

ultrasonic-frequency pulsed arc mainly affects the arc and molten pool, including arc constriction, heat-flux redistribution, and molten-pool flow. However, its direct action on the solidification front is relatively weak. Mechanical vibration, in contrast, is more closely related to solidification-stage regulation, but its effect is limited in dense metallic melts because of acoustic attenuation and the difficulty of cavitation activation. Their combination therefore provides a more continuous regulation route from molten-pool flow to solidification. Second, the pulsed arc also modifies the thermal state and thermophysical properties of the molten pool. The hotter and less viscous melt responds more effectively to the mechanical vibration transmitted through the plate. Therefore, HU-TIG is not a simple addition of two ultrasonic inputs, but a coupled process in which the pulsed arc improves both molten-pool flow and the effectiveness of mechanical ultrasonic action during solidification.

At the process level, arc constriction and molten-pool heating improve heat-transfer efficiency and strengthen Marangoni convection. The decreases in viscosity, surface tension, and acoustic attenuation also reduce the loss of acoustic energy in the melt and make cavitation more likely once sufficient rarefaction pressure is reached. Under this condition, acoustic streaming and cavitation-related loading may develop more effectively near the solidification front. During solidification, the enhanced flow promotes heat and solute transport, while the reduced thermal gradient and the wider constitutional-supercooling zone increase the tendency for CET. Acoustic loading near the solidification front may further fragment dendrite arms and provide additional nucleation sites. At the same time, solute redistribution and defect generation favor intragranular austenite precipitation, suppress the dominance of Widmanstätten growth, and reduce K–S-related α/γ interfaces.

These microstructural changes are closely related to the tensile behavior. Ferrite grain refinement contributes to the increase in strength. The more uniform ferrite/austenite distribution and fewer continuous K–S-related interfaces reduce local strain concentration and make crack propagation less continuous. This is consistent with the simultaneous improvement in tensile strength and elongation, as well as the more ductile fracture morphology observed in HU-TIG. The results also indicate that the molten pool is not simply a passive medium for ultrasonic transmission. Its thermal and physical state affects how efficiently the external vibration is transmitted and how strongly it can act during solidification. In this sense, controlling the acoustic response of the molten pool may be important for ultrasonic-assisted welding and melt-based additive manufacturing of dense alloys.

5. Conclusions

This work demonstrates that coupling arc-induced ultrasonic with externally applied mechanical vibration provides a simple and effective approach for precise microstructure control in welds. The significant optimization of microstructure and mechanical properties achieved in the 2507 SDSS welds validates the feasibility of this method; moreover, mechanistic analysis of the solidification thermodynamics and acoustic effect provides a theoretical foundation for its extension to other alloys. The main conclusions are summarized as follows:

1. The hybrid strategy significantly optimizes weld geometry, increasing penetration by 100.7% relative to C-TIG. This improvement is driven by the combined enhancement of Marangoni convection and ultrasonic-induced streaming, which collectively promote heat and mass transfer within the molten pool.
2. The hybrid ultrasonic strategy improves melt acoustic responsiveness compared with single-mode counterparts. The ultrasonic-

frequency pulsed arc elevates the molten pool temperature, thereby reducing viscosity and surface tension, weakening viscous dissipation, and facilitating acoustic energy transmission. Under these thermo-physical conditions, the effectiveness of ultrasonic agitation is enhanced, and cavitation becomes more feasible when local pressure falls below the cavitation threshold.

3. A simultaneous improvement in strength and ductility was achieved: ferrite grain size was refined by ~51% (from 143.4 μm to 70.2 μm), leading to a 97% increase in elongation and a 16% rise in UTS. This refinement is attributed to the synergistic effect of altered solidification thermodynamics and intensified ultrasonic effect.
4. The hybrid ultrasonic strategy promotes austenite formation and optimizes phase interface characteristics: the austenite area fraction increases from 6.9% to 20.0%, while the K-S boundary fraction decreases from 38.6% to 20.3%. The underlying mechanisms include enhanced molten pool convection and increased nucleation sites for intragranular austenite.

Overall, this work not only proposes a hybrid ultrasonic strategy but also enhances the effect of ultrasonic field on the melting-solidification process by tuning the acoustic responsiveness of the melts. The mechanistic insights into the fundamental mechanisms of this strategy can be generalized to the welding, repair, and additive manufacturing of other alloys such as titanium alloys and nickel-based superalloys. Built on the hybrid ultrasonic strategy and the concept of acoustic responsiveness tuning, this microstructural control approach offers a promising pathway to address longstanding challenges in precision manufacturing and the fabrication of high-performance structural components.

In future work, we will extend this hybrid strategy to arc-based directed energy deposition and systematically investigate the evolution of microstructure and mechanical properties under repeated remelting and multiple thermal cycling conditions. This will further validate the versatility and robustness of the strategy in complex thermal processing scenarios and expand its application scope to additive manufacturing.

CRediT authorship contribution statement

Qihao Chen: Software. **Sanbao Lin:** Supervision, Project administration. **Yangyang Fan:** Conceptualization. **Bolun Dong:** Writing – review & editing. **Xiaoyu Cai:** Resources. **Yunhao Xia:** Writing – original draft, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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